Math GRE Prep Day 3

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Outline

1. Calculus Problems
   - Somewhat trickier problems

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Problem 11

Evaluate

$$\lim_{n\to\infty}\left(\sqrt{1+2+3+\cdots+n}-\sqrt{1+2+3+\cdots+(n-1)}\right)$$

Problem 12

Evaluate

$$\lim_{n\to\infty} \frac{\sqrt{1\cdot 2}+\sqrt{2\cdot 3}+\cdots+\sqrt{n(n+1)}}{n^2}$$

Problem 13

Evaluate
$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 14

Evaluate

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{3}.$$

Problem 15

If $f(x) = e^{x^3} + e^{-x^3}$, evaluate $f^{(2022)}(0)$

Problem 16

Calculate

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{\sqrt{1^3 + 2^3 + \cdots + i^3}}$$

Problem 17

Calculate

$$\lim_{x\to\infty}\left(\sin\sqrt{x+\pi}-\sin\sqrt{x}\right)$$

Problem 18

Calculate

$$\lim_{n\to\infty} \cos\frac{1}{2}\cos\frac{1}{4}\cos\frac{1}{8}\dots\cos\frac{1}{2^n}$$